# Hover or Crash — Design Report

Drone Motor Selection Design Challenge · EE 250 · Instructor: Bhaskar Krishnamachari
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## Flight crew

Name: Leo
Teammates: invisible george and invisible elizabeth

## Mission

Campus Rooftop Inspection: 1 lb quad (airframe 300 g + 4 motors), 3S Li-Po 11.1 V 1500 mAh, 13.4 in wheelbase so adjacent motors are 9.5 in apart (props wider than that collide), hover 10 min with one gust, budget $200 for the motor set.

## Sizing lab (student entries vs. model)

| Quantity | Student | Model | Unit |
|---|---|---|---|
| Total weight force | 4.448 | 4.448 | N |
| Hover thrust per motor | 1.112 | 1.112 | N |
| Design target per motor (2x) | 2.224 | 2.224 | N |
| Rotational speed n | 103.2 | 103.2 | rev/s |
| rpm | 6192 | 6192 | rpm |
| Target KV = 2 x rpm / V (8 in prop, 3S) | 1116 | 1116 | rpm/V |

## Scorecard (1–5, auto-rated by the simulator)

| Criterion | A: 2300KV 5×3 | B: 1500KV 7×4 | C: 920KV 9×4.5 | D: 600KV 10×4.5 |
|---|---|---|---|---|
| Meets thrust target | 5 | 5 | 5 | 1 |
| Thrust-to-weight | 4 | 5 | 5 | 1 |
| Cost | 5 | 4 | 2 | 2 |
| Agility / response | 5 | 4 | 3 | 2 |
| Frame fit / clearance | 5 | 5 | 3 | 1 |
| Endurance (10 min) | 2 | 4 | 5 | 5 |
| Total (of 30) | 26 | 27 | 23 | 12 |
| Sim endurance | 8:34 | 12:54 | 13:47 | n/a |
| Thrust : weight | 7.3:1 | 8.7:1 | 9.1:1 | 5.6:1 |

## Test flights flown

Option A — FORCED LANDING. Battery empty at 8:34; the mission needed 10:00. Spinning 5-inch props at 2300 KV takes about 117 W just to hover, and a 1500 mAh 3S pack (16.7 Wh) can't supply that for 10 minutes.
Option B — MISSION COMPLETE. Clean hover, recovered the gust, landed with reserve.
Option C — MISSION COMPLETE. Prop tips pass within 0.25 in of each other; any arm flex could cause a strike. Recovery from the gust was slow (large prop inertia).
Option D — CRASH. Prop strike. Adjacent motors are 9.5 in apart, so two 10-inch props overlap by 0.5 in and hit

each other on spool-up.
Option B — MISSION COMPLETE. Clean hover, recovered the gust, landed with reserve.

## Student decision

### Recommended option: B — 1500 KV, 7×4 prop, $125 per set

**Why I chose it:**
the appropriate prop size margin

**Why I did not choose the others:**
cost + inappropriate prop size

## Simulator feedback

• Option B meets the thrust target with 8.7:1 thrust-to-weight and 12:54 of endurance, so it completes the mission.
• This matches the simulator's top in-budget option (27/30).
• Options flown: A, B, C, D.
• KV check: your 8-inch target was ~1116 rpm/V; scaled by (8/D)² for a 7-inch prop that is ~1499, against a catalog 1500. The rule puts the needed rpm at half the motor's top speed (KV × V), leaving the other half for gusts and battery sag.

*Model notes: thrust from T = Ct·rho·n²·Dt with Ct = 0.10; endurance from a per-option efficiency*
*inertia. These are teaching approximations, not measured data.*